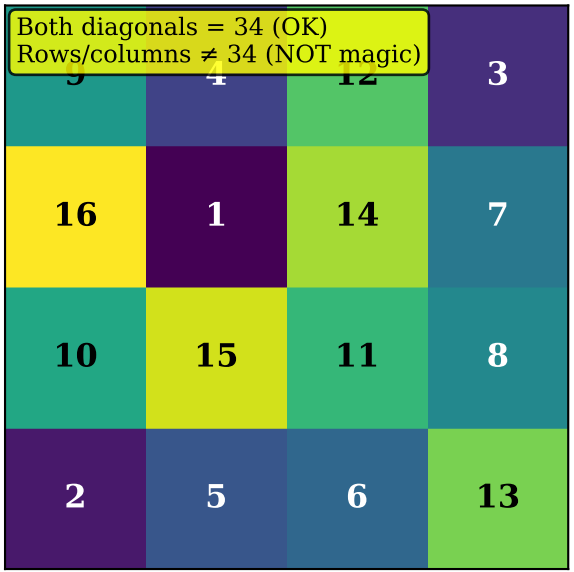
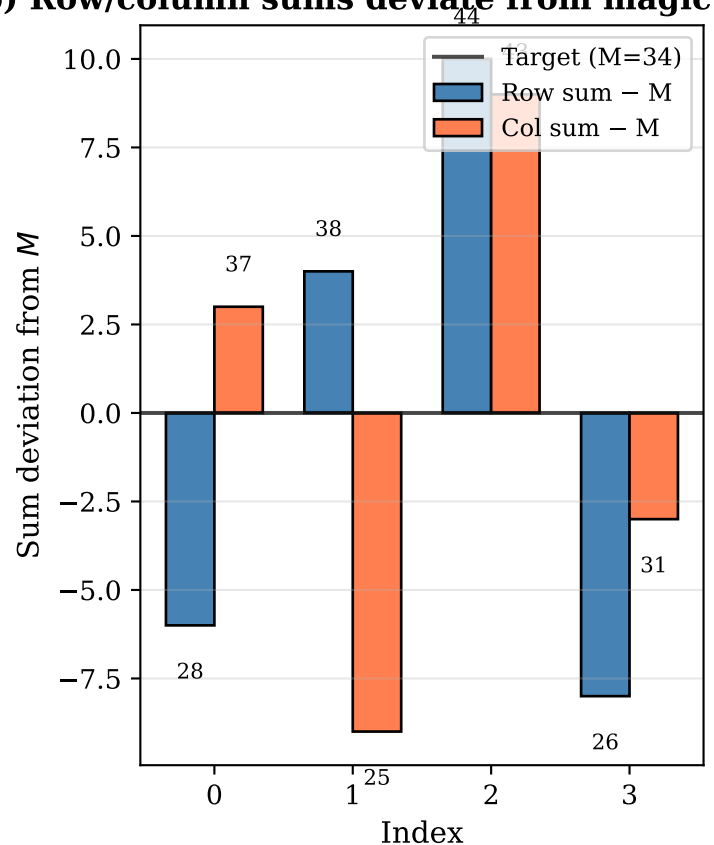
Why E_{low} Cannot Characterize Magic Squares for $n \geq 4$

(a) Example: $E_{\text{low}} = 0$ but NOT magic



(b) Row/column sums deviate from magic constant



(c) Low-mode energy vanishes

Constraint	Value
$\text{Cov}(X, Z)$	0.00e+00
$\text{Cov}(Y, Z)$	0.00e+00
$\text{Cov}(D_{\text{main}}, Z)$	0.00e+00
$\text{Cov}(D_{\text{anti}}, Z)$	0.00e+00
E_{low}	**0.00e+00**

All four covariances = 0, yet NOT magic!